



Data Analysis Toolkit

Complete Documentation, User Guide & FAQ

v1.0	8 Modules	Browser-Based	No Installation
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8

Analysis Modules

50+

Statistical Functions

0

Lines of Code Needed

100%

Browser-Based

All calculations run locally in your browser. No data is sent to any server.

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1. Getting Started

The Data Analysis Toolkit is a fully self-contained, browser-based statistical workbench. It requires no installation, no internet connection after loading, and no programming knowledge. Simply open the file and start analysing.

System Requirements

Requirement	Details
Browser	Chrome 90+, Firefox 88+, Edge 90+, Safari 14+
Internet	Only needed for first load (fonts + Chart.js CDN). Works offline after.
Files	Single HTML file — index.html (approx. 75 KB)
Data format	CSV or TSV files. UTF-8 encoding recommended.
Screen size	Optimised for desktop (1024px+). Usable on tablet.

Quick Start (3 Steps)

Step 1	Open index.html Double-click the file or drag it into any modern web browser. The toolkit loads with the built-in sample dataset (20 rows, 8 variables) automatically — no setup needed.
Step 2	Load your data (optional) Click the blue "Load CSV" button in the top-right header. Select any comma-separated file. Column types are detected automatically and all 8 modules update instantly.
Step 3	Navigate the modules Click any tab in the navigation bar to switch between modules. Each module is independent — changes in one do not affect others.

Tip: The built-in sample dataset contains 20 participants with age, height, weight, test score, city, gender, and income — ideal for trying every feature before loading your own data.

2. Interface Overview

The interface is organised into three fixed zones that are always visible regardless of which module is active.

Zone	Description
Header Bar	Sticky bar at the top of the page. Contains the toolkit logo, title, and the three global action buttons: Load CSV, Export, and Auto-Insights. This bar scrolls with you.
Navigation Bar	Scrollable tab strip below the header. Contains one tab per module. The active tab is highlighted in blue with an underline. On smaller screens, the tab bar scrolls horizontally.
Content Area	The main working area. Displays the currently active module. Each module is divided into cards — white panels with a title and interactive controls.

Common UI Elements

Dropdown (select): Click to open a list of options. Used for choosing columns, distributions, and methods throughout the toolkit.

Range slider: Drag left/right to adjust a numeric value. The current value is always shown next to the slider in monospace font.

Toggle buttons: A group of labelled buttons where exactly one is active (shown in blue). Clicking a different button switches the active selection.

Card: A bordered panel with a grey title and white background. Each card groups related controls and outputs together.

Result box: Coloured panel that appears after a calculation. Green border = pass/success, Red border = reject/fail, Amber border = warning.

Badge: Small coloured pill label. Blue = numeric type, Purple = categorical type, Green = no issues, Red = missing/outlier.

Key-value row: A label on the left with a monospace value on the right, separated by a thin line. Used to display statistics.

3. Header Buttons Reference

Three global buttons appear in the header at all times. They operate on the entire dataset and are accessible from any module.

Button	Label	Colour	What it does	When to use
Load CSV	Load CSV	Blue (primary)	Opens a file-picker dialog. Select any .csv or .tsv file. The toolkit parses it, auto-detects column types, populates all module dropdowns, and switches to the Explorer tab.	Whenever you want to analyse your own data instead of the built-in sample.
Export	Export	Grey (secondary)	Downloads the current dataset (including any imputed values from the Cleaning module) as a CSV file named toolkit_export.csv.	After cleaning or imputing your data and wanting to save the processed version.
Auto-Insights	Auto-Insights	Grey (secondary)	Switches to the Data Story tab and runs the automatic insight generator, which scans every column for skewness, outliers, missing values, correlations, and normality.	As a first step after loading new data to get a quick narrative summary of what is in the dataset.

Note: "Load CSV" replaces the entire dataset. Any imputation changes made in the Cleaning tab will be lost. Use Export first if you want to keep them.

4. Module 1 — Data Explorer

The Explorer is the entry point for every analysis. It gives you a complete structural summary of your dataset before you begin any calculations. It loads automatically when the toolkit opens.

Summary Metric Cards

Five metric cards are displayed at the top of the Explorer. They update instantly when a new CSV is loaded.

Metric Card	Colour	What it shows
Rows	Blue	Total number of data rows in the dataset (header not counted).
Columns	Green	Total number of columns detected.
Missing	Red	Total count of null/empty/NA cells across the entire dataset.
Numeric	Purple	Number of columns classified as numeric ($\geq 70\%$ numeric values).
Categorical	Amber	Number of columns classified as categorical (text values).

Missing Value Analysis Panel

For every column, a horizontal bar shows what percentage of values are missing. A green bar means zero missing values. A red bar indicates the proportion that is null. The exact count and percentage are shown beside each bar.

Column Schema Table

Column	What it shows
Column name	Name of the column as read from the CSV header row.
Type badge	Blue "numeric" or Purple "categorical" — the auto-detected data type.
Unique	Count of distinct non-null values in the column.
Missing badge	Green "0" if complete, Red count if any nulls exist.

Data Preview Table

Displays the first 10 rows of the dataset in a scrollable table. Null values are shown as a red "null" badge. This confirms the CSV was parsed correctly before running any analysis.

5. Module 2 — Descriptive Statistics

Computes a full numerical summary for any numeric column. Select the column from the dropdown and all statistics update instantly. Use the "Full summary table" button to see all numeric columns side by side.

Controls

Control	Type	What it does
Select column dropdown	Dropdown	Choose which numeric column to summarise. All statistics panels and the histogram update immediately.
Full summary table	Button	Shows/hides a wide table comparing all numeric columns at once — n, mean, median, SD, min, max, skewness, kurtosis.

Statistics Reference

Statistic	Symbol	Formula / Definition	Typical range
Count	n	Non-null observations	Integer ≥ 0
Mean	x-bar	Sum of values divided by n	Any real number
Median	M	50th percentile (middle value)	Any real number
Std Deviation	s	Square root of sample variance	≥ 0
Variance	s^2	Sum of squared deviations / (n-1)	≥ 0
Minimum	min	Smallest observed value	Any real number
Q1 (25th)	Q1	25th percentile	Any real number
Q3 (75th)	Q3	75th percentile	Any real number
Maximum	max	Largest observed value	Any real number
IQR	IQR	$Q3 - Q1$	≥ 0
Skewness	g1	Fisher's moment coefficient (0 = symmetric)	Typically -3 to +3
Kurtosis	g2	Excess kurtosis (normal = 0)	Typically -2 to +10
CV	CV%	$(SD / \text{Mean}) \times 100$	> 0 (higher = more spread)
SEM	SE	$SD / \sqrt{n}$	≥ 0

Five-Number Summary Visual

Below the statistics table, a horizontal strip visualises the five-number summary. The blue shaded region covers the IQR (Q1 to Q3). The vertical blue line marks the median. The five values — min, Q1, median, Q3, max — are printed below the strip.

Frequency Distribution Table

The data is divided into 7 equal-width bins. For each bin the table shows: bin range, count, relative frequency (%), and cumulative frequency (%).

Interpretation Guide

Condition	Interpretation
Skewness > +1	Right-skewed: long tail to the right. Mean > Median. Common in income, prices.
Skewness < -1	Left-skewed: long tail to the left. Mean < Median.
Skewness < 0.5	Approximately symmetric distribution.
Kurtosis > 2	Heavy tails — more outliers than a normal distribution.
Kurtosis < -2	Light tails — fewer extreme values than normal.
CV > 30%	High relative variability. The spread is large compared to the average.
Mean != Median	Likely skewed or outliers present. Median is more robust.

6. Module 3 — Visualizations

Provides five interactive chart types. Select a chart type using the toggle buttons, choose columns from the controls, and the chart updates instantly. Two supplementary charts (category distribution and group comparison) are always shown below the main chart.

Chart Type Buttons

Button	Chart Type	Required Inputs	Best used for
Histogram	Bar chart of frequency bins	One numeric column + bin count slider	Seeing the shape, spread, and mode of a single variable.
Scatter	Point plot of two variables	X column + Y column (both numeric)	Detecting correlations and relationships between two numeric variables.
Box plot	IQR box with whiskers	One numeric column	Comparing spread, quartiles, and outlier candidates across columns.
Bar chart	Frequency bar chart	One categorical column	Counting how often each category appears (e.g. city, gender).
Correlation matrix	Colour-coded heatmap grid	All numeric columns (auto)	Getting a quick overview of which pairs of numeric variables are correlated.

Per-Chart Controls

Histogram

Column dropdown: Select which numeric column to plot.

Bins slider: Drag to set the number of histogram bins (3 to 20). Fewer bins = coarser view; more bins = finer view. Current value shown beside slider.

Scatter Plot

X dropdown: Variable plotted on the horizontal axis.

Y dropdown: Variable plotted on the vertical axis.

Chart title: Automatically updated to show "Scatter: X vs Y (r = value)" where r is the Pearson correlation coefficient, computed live.

Box Plot

Column dropdown: Select the numeric variable to summarise. The chart shows three coloured segments: min-to-Q1 (grey), Q1-to-Q3 (blue IQR box), and Q3-to-max (grey).

Bar Chart

Category dropdown: Select a categorical column. Each unique category becomes a bar whose height equals its row count.

Correlation Matrix

No manual controls — uses all numeric columns automatically (up to 6). Each cell colour represents the Pearson r between the row and column variables: dark blue = strong positive, dark red = strong negative, transparent = near zero. Hover over a cell to see the exact r value.

Supplementary Charts

Chart	What it shows
Category distribution	Doughnut chart of the first categorical column. Always shows the proportional breakdown by category.
Group comparison	Bar chart comparing the mean of the first numeric column across groups defined by the first categorical column.

Reading a correlation matrix: $r > 0.7$ = strong positive, r 0.4-0.7 = moderate, r 0.1-0.4 = weak, r near 0 = no linear relationship. Negative values indicate inverse relationships.

7. Module 4 — Probability Distributions

Provides interactive visualisation of six probability distributions plus a goodness-of-fit test. Use the distribution buttons to switch families, then drag the parameter sliders to see the PDF and CDF update in real time.

Distribution Buttons

Button	Distribution	Parameters	Key properties
Normal	Gaussian bell curve	mu (mean), sigma (std dev)	Symmetric, defined on all reals. Most common in statistics.
Exponential	Decay curve	lambda (rate)	Right-skewed, models waiting times. Always ≥ 0 .
Uniform	Flat rectangle	a (min), b (max)	All values equally likely in [a, b]. Zero outside.
Binomial	Discrete bar chart	n (trials), p (prob)	Counts successes in n independent trials. Integer values only.
Poisson	Discrete bar chart	lambda (rate)	Counts events in a fixed interval. Integer values only.
Student t	Bell curve (heavier tails)	df (degrees of freedom)	Like Normal but with heavier tails. Used for small samples.

Parameter Sliders

Each distribution shows one or two sliders, one per parameter. Drag the slider to change the parameter value. The current value is displayed in monospace beside the label. Both the PDF chart and CDF chart update immediately.

Parameter	Appears in	Effect when increased
mu (mean)	Normal	Shifts the bell curve left or right along the x-axis.
sigma (std dev)	Normal	Widens the bell curve (more spread). Narrower curve when decreased.
lambda (rate)	Exponential, Poisson	Exponential: faster decay, smaller mean. Poisson: higher expected count.
a (min)	Uniform	Moves the left edge of the rectangle.
b (max)	Uniform	Moves the right edge; must be $> a$.
n (trials)	Binomial	More bars, distribution becomes more symmetric (approaches Normal).
p (prob)	Binomial	Shifts the peak right (higher p = more successes).
df	Student t	Heavier tails at low df. Approaches Normal as df $\rightarrow$ infinity.

Distribution Statistics Row

Below the PDF/CDF charts, four metric cards display the theoretical moments of the current distribution: Mean, Variance, Skewness, and Excess Kurtosis. These update with the sliders.

Goodness-of-Fit Section

Tests whether your data is consistent with a chosen theoretical distribution using the Kolmogorov-Smirnov (KS) test.

Control / Output	Type	Description
Column dropdown	Dropdown	The numeric variable from your dataset to test.
Distribution dropdown	Dropdown	The theoretical distribution to test against (Normal, Exponential, or Uniform).
D statistic	Output	Maximum absolute difference between the empirical CDF and the theoretical CDF. Smaller = better fit.
D critical	Output	Threshold at $\alpha=0.05$. Calculated as $1.36 / \sqrt{n}$.
Verdict badge	Output	Green "Fits [dist]" if $D \leq D_{critical}$. Red "Does not fit" otherwise.
Q-Q Plot	Chart	Theoretical quantiles (x-axis) vs sample quantiles (y-axis). Points along the diagonal line = good fit to Normal.

8. Module 5 — Statistical Inference

Four inferential tools arranged in a 2x2 grid: confidence interval, one-sample t-test, two-sample t-test, and normality test. All update automatically when a control changes.

Card 1 — Confidence Interval

Control / Output	Type	Description
Column dropdown	Dropdown	Numeric variable whose mean you want to estimate.
Confidence level slider	Slider (80-99)	Drag to set the level. Higher confidence = wider interval. Default: 95%.
Level display	Label	Shows current level as "XX%". Updates live with the slider.
Interval [lower, upper]	Output	The computed confidence interval in large blue monospace text.
x-bar	Output	Sample mean.
s (std dev)	Output	Sample standard deviation.
SE (standard error)	Output	SD / sqrt(n). The precision of the mean estimate.
t* (critical value)	Output	The t critical value for the chosen level and df.
Margin of error	Output	t* times SE. Half the interval width.
n	Output	Sample size (non-null values).

Card 2 — One-Sample t-Test

Control / Output	Type	Description
Column dropdown	Dropdown	The numeric variable to test.
H0: mu = slider	Slider	Set the hypothesised mean. Drag to any value. Range auto-set based on column max.
Alpha dropdown	Dropdown	Significance level: 0.01, 0.05 (default), or 0.10.
t statistic	Output	Computed as $(\bar{x} - \mu_0) / SE$. Measures how many standard errors the sample mean is from the null.
df	Output	Degrees of freedom = $n - 1$.
p-value	Output	Two-tailed probability of observing this t or more extreme under H0. Approximated using normal.
Decision badge	Output	Red "Reject H0" if $p < \alpha$. Green "Fail to reject H0" otherwise.
Interpretation text	Output	Plain-English statement of the conclusion.

Card 3 — Two-Sample t-Test (Welch)

Compares the means of two groups using Welch's t-test, which does not assume equal variances (more robust than the classic Student t-test for two samples).

Control / Output	Type	Description
Column dropdown	Dropdown	The numeric outcome variable to compare between groups.
Group by dropdown	Dropdown	The categorical variable that defines the two groups (e.g. gender -> M vs F).
Group 1 / Group 2 mean	Output	Mean and sample size for each of the first two unique group values.
t statistic	Output	Welch t: $(\text{mean1} - \text{mean2}) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$.
df (Welch)	Output	Degrees of freedom from the Satterthwaite approximation (decimal number).
Cohen's d	Output	Effect size using pooled SD. Shown with magnitude label: negligible/small/medium/large.
Decision badge	Output	Red "Reject H0" or Green "Fail to reject" at $\alpha=0.05$.

Card 4 — Normality Test

Control / Output	Type	Description
Column dropdown	Dropdown	Select the numeric variable to test for normality.
Skewness	Output	Sample skewness with badge: Green "OK" if $ \text{skew} < 1$, Amber "notable" if ≥ 1 .
Excess kurtosis	Output	Excess kurtosis with badge: Green "OK" if $ \text{kurt} < 2$, Amber "notable" otherwise.
Jarque-Bera	Output	JB statistic and p-value. Tests if skewness and kurtosis jointly match normal.
Shapiro-Wilk W	Output	Approximated W statistic. Green "near normal" if $W > 0.9$, Red "concern" if < 0.9 .
Verdict badge	Output	Green "Approximately normal" or Amber "May deviate" based on all three tests.

p-value quick reference: $p < 0.001$ = very strong evidence; 0.001-0.01 = strong; 0.01-0.05 = moderate (reject at $\alpha=0.05$); 0.05-0.10 = weak/marginal; ≥ 0.10 = little evidence against H0.

9. Module 6 — Regression

Fits a simple ordinary-least-squares (OLS) linear regression of one variable on another. The model updates whenever you change the X or Y dropdowns.

Column Selection

Control	Type	Description
X (predictor) dropdown	Dropdown	The independent variable (input). Placed on the x-axis of the scatter plot.
Y (response) dropdown	Dropdown	The dependent variable (outcome). Placed on the y-axis. Should differ from X.

Model Outputs

Output	Symbol	Description
Model equation	$\hat{y} = bx + a$	The fitted line equation displayed in large cyan monospace text. b is slope, a is intercept.
R (correlation)	r	Pearson correlation coefficient. Sign shows direction; magnitude shows strength.
R-squared	R^2	Proportion of Y variance explained by X. Range 0-1. Higher is better.
Adjusted R-squared	Adj- R^2	R^2 penalised for sample size. More honest for small datasets.
RMSE	RMSE	Root mean squared error — average prediction error in the original Y units.
F statistic	$F(1, n-2)$	Tests overall model significance. Large F with small p = model is useful.
n	n	Number of complete (non-null) pairs used.

Coefficient Table

Column	Description
Term	Intercept or the name of the predictor column.
Estimate	The fitted coefficient value (a for intercept, b for slope).
SE	Standard error of the coefficient estimate.
t	t-statistic = Estimate / SE. Tests if the coefficient is significantly different from zero.
p-value	Two-tailed p-value. Green badge if $p < 0.05$ (significant), Amber if not.

Charts

Scatter with regression line: Blue points show the raw data. Red line shows the fitted OLS line. The line extends from the minimum to maximum observed X value.

Residuals vs fitted: Purple points. X-axis = fitted (predicted) values; Y-axis = residuals (actual minus predicted). A red horizontal line at $y=0$ is the reference. Ideally, points should scatter randomly around zero with no pattern.

Predict Panel

Control / Output	Type	Description
x = input field	Number input	Type any numeric value for the predictor variable X.
Predict button	Button	Computes the predicted Y value and a 95% prediction interval.
y-hat	Output	Point prediction: $a + b * x_{\text{new}}$, shown in large cyan text.
95% prediction interval	Output	Range [lower, upper] where a new individual observation is expected to fall with 95% probability.

Assumption check: Look at the residuals vs fitted plot. If you see a fan shape (variance increases with fitted values), the model may violate the constant-variance assumption. Random scatter = good.

10. Module 7 — Probability Calculator

Computes exact probabilities for any of four supported distributions. Choose the distribution, set parameters, choose the query type, enter the value(s), and click Calculate.

Setup Controls

Control	Type	Options / Range	Description
Distribution dropdown	Dropdown	Normal, Exponential, Uniform, Student t	Selects the probability distribution. Parameter sliders below update to match.
Parameter sliders	Sliders	Vary by distribution	Set distribution parameters (same as Distributions module). Chart updates live.
Query type dropdown	Dropdown	4 options (see below)	Chooses what probability question to ask.

Query Types

Query type	Input needed	Output	Example use
$P(X \leq x)$	One x value	Left-tail CDF probability	What fraction of students scored below 80?
$P(X \geq x)$	One x value	Right-tail probability = $1 - \text{CDF}(x)$	What fraction scored above 80?
$P(a \leq X \leq b)$	Two values: a and b	Interval probability = $\text{CDF}(b) - \text{CDF}(a)$	What fraction scored between 70 and 90?
Inverse CDF	A probability p (0-1)	x such that $P(X \leq x) = p$	What score separates the bottom 25%?

Calculate Button

After setting the distribution, parameters, and query type, click the blue "Calculate" button. The result appears immediately in a green result box showing the probability equation and its value to 6 decimal places.

Distribution Curve Chart

The right panel always shows the PDF of the current distribution with the current parameters. For discrete distributions (Binomial, Poisson), a bar chart is shown. For continuous, a filled line chart. The chart updates whenever any slider or dropdown changes.

Example: Normal distribution, $\mu=75$, $\sigma=10$. Query: $P(X \leq 80)$. Answer: 0.691462. This means about 69.1% of the distribution is below 80.

11. Module 8 — User Guide (Built-in)

The last tab in the navigation bar is a built-in User Guide rendered directly inside the toolkit. It provides quick-reference summaries for each module, a p-value interpretation table, and key formulas — all accessible without leaving the app. This PDF is the extended version of that guide.

12. Complete Worked Example

This section walks through a full analysis of the built-in sample dataset step by step, demonstrating every module with real inputs and expected outputs.

The Dataset

The built-in dataset contains 20 participants with 8 variables. Three values are intentionally missing (weight rows 6 and 13; score rows 9 and 17).

Variable	Type	Description	Example values
id	Numeric	Row identifier	1, 2, 3, ...
age	Numeric	Age in years	22 to 55
height	Numeric	Height in cm	158 to 183
weight	Numeric	Weight in kg (3 missing)	52 to 92
score	Numeric	Test score 0-100 (3 missing)	59 to 95
city	Categorical	City name	Berlin, Hamburg, Munich, Frankfurt
gender	Categorical	M or F	M, F
income	Numeric	Annual income in EUR	28000 to 72000

Step-by-Step Walkthrough

Step 1	Open the toolkit [Explorer tab]
Open index.html in your browser. You will see 6 metric cards: Rows=20, Columns=8, Missing=4, Numeric=6, Categorical=2, Completeness=97%.	
Expected result: Confirm: The missing bar for "weight" should show red at 15%.	
Step 2	Check the schema [Explorer tab]
In the Column schema table, verify: age/height/weight/score/income all show blue "numeric" badges. City and gender show purple "categorical" badges.	
Expected result: If any column is misclassified, check your CSV for mixed data types in that column.	

Step 3	Explore income statistics [Descriptive Stats tab]
	Select "income" from the column dropdown. Expected: Mean ~51,850; Median ~51,500; SD ~13,800; Skewness ~0.15 (approximately symmetric); CV ~26.6%.
	<i>Expected result: Since $CV < 30\%$ income has moderate variability. Skewness near zero confirms a roughly symmetric distribution.</i>
Step 4	Compare income by gender [Visualizations tab]
	Click "Scatter", set X = age, Y = income. You should see a positive trend with r approximately +0.77, meaning older participants tend to earn more in this sample.
	<i>Expected result: Then click "Correlation matrix" to see that age-income is the strongest pair at $r \sim 0.77$.</i>
Step 5	Test if mean score = 80 [Inference tab]
	Select "score" in the one-sample t-test card. Drag the H0 slider to 80. Expected: t ~ -0.8, p ~ 0.44. Decision: Fail to reject H0.
	<i>Expected result: This means there is no significant evidence that the true mean score differs from 80 at $\alpha=0.05$.</i>
Step 6	Build a confidence interval for age [Inference tab]
	Select "age" in the CI card. Set level to 95%. Expected: approximately [33.0, 41.5] meaning we are 95% confident the true mean age is in this range.
	<i>Expected result: Lower the level to 80% and observe the interval narrows. Raise to 99% and it widens.</i>
Step 7	Fit Normal to score [Distributions tab]
	In the Goodness-of-Fit section, select column = score, distribution = Normal. Expected: D ~ 0.10-0.15, D_critical ~ 0.32 (n=17). Verdict: Fits Normal.
	<i>Expected result: The Q-Q plot should show points roughly along the diagonal line, confirming approximate normality.</i>

Step 8	Regress income on age [Regression tab]
Set X = age, Y = income. Expected: slope ~ 870 (each extra year adds ~870 EUR), $R^2 \sim 0.60$, F significant ($p < 0.001$).	
<i>Expected result: Use the Predict panel: type $x = 35$. Expected prediction: ~43,000 EUR. The 95% interval will be wide due to small n.</i>	
Step 9	Compute P(score >= 85) [Probability Calculator tab]
Set distribution = Normal, $\mu = 80$, $\sigma = 10$. Query = $P(X \geq x)$, $x = 85$. Expected: ~0.3085 (about 30.9% of the distribution).	
<i>Expected result: Switch query to Inverse CDF, enter $p = 0.90$. Result: $x \sim 92.8$. This is the 90th percentile of this Normal distribution.</i>	
Step 10	Run Auto-Insights [Data Story tab]
Click the "Auto-Insights" button in the header. Review the generated insights. Expect to see: correlation flag for age-income, skewness check for income, missing value warnings for weight and score, and outlier scan.	
<i>Expected result: This is a good sanity check after loading any new dataset.</i>	

13. Frequently Asked Questions (FAQ)

General

Q Does the toolkit require an internet connection?

Only for the very first load, when it fetches the Inter and JetBrains Mono fonts from Google Fonts and Chart.js from cdnjs. After the first load, the browser caches these resources. To use fully offline, download Chart.js locally and host the fonts locally, then update the script and link tags in the HTML head.

Q Is my data sent to any server?

No. All data processing happens entirely in your browser using JavaScript. No CSV data, no statistics, and no results are ever transmitted anywhere. You can disconnect from the internet after loading the page and the toolkit will continue to work.

Q Can I use this on a tablet or phone?

The toolkit is designed for desktop screens (1024px or wider). On tablets in landscape mode it is usable. On phones, the content is very cramped. The navigation bar scrolls horizontally on small screens.

Q How do I reset back to the sample dataset?

Refresh the page (F5 or Cmd+R). The toolkit always starts with the built-in sample dataset.

Q Can I save my work?

Use the "Export" button to download the current dataset as a CSV. All chart images can be saved by right-clicking on them and selecting "Save image as". There is no session persistence — refreshing the page resets all state.

Data Loading

Q What file formats are supported?

CSV (comma-separated values) and TSV (tab-separated values). The file must have a header row as the first line. UTF-8 encoding is strongly recommended. The parser also handles quoted fields containing commas.

Q What counts as a missing value?

The following values are treated as null: empty cell, NA, N/A, null, NULL, NaN, and ?. Any other non-numeric string in a numeric column is also treated as null.

Q My column types are wrong after loading. Why?

A column is classified as numeric if at least 70% of its non-null values parse as numbers. If a mostly-numeric column contains text labels or units (like "35kg"), it will be classified as categorical. Clean the column in your spreadsheet before loading, or use the filter controls in the Inference/Regression tabs which operate only on non-null numeric values.

Q How many rows and columns can the toolkit handle?

Practically, it works well up to about 5,000 rows and 20 columns. Beyond that, chart rendering may slow down. There is no hard limit, but very large datasets will make the browser tab feel sluggish.

Q My CSV has special characters and loads incorrectly.

Save your file as UTF-8 in Excel (File > Save As > CSV UTF-8) or LibreOffice. Avoid using semicolons as delimiters — the parser expects commas. If your values contain commas, ensure they are enclosed in double quotes in the CSV.

Statistics Questions

Q What is the difference between standard deviation and standard error?

Standard deviation (s) measures the spread of individual data points around the mean. Standard error of the mean ($SEM = s / \sqrt{n}$) measures the precision of the mean estimate itself — how much the sample mean would vary if you repeated the study. SEM is always smaller than SD and decreases as n increases.

Q Why does the CI get wider when I increase the confidence level?

A higher confidence level (e.g. 99% vs 95%) means we want to be more certain that the interval captures the true mean. To be more certain, we must cast a wider net. Mathematically, the critical t -value t^* increases with the confidence level, which directly widens the margin of error.

Q The two-sample t-test says "Fail to reject H0" but the means look different on the chart. Why?

Statistical significance depends on both the size of the difference AND the sample size. With only 20 rows split across two groups (roughly 10 each), the test has low statistical power — it can only detect large differences reliably. The same difference with $n=200$ per group would likely be significant.

Q Why does the Shapiro-Wilk W show a note saying it's an approximation?

The true Shapiro-Wilk test requires exact coefficient tables for each sample size. This toolkit computes an approximation using the correlation between sample quantiles and theoretical normal quantiles (the basis of the Q-Q plot). For practical teaching purposes the result is a reliable indicator, but for formal publication use R or Python `scipy.stats.shapiro`.

Q What does R-squared of 0.65 mean in regression?

$R^2 = 0.65$ means that 65% of the variance in the Y variable is explained by the linear relationship with X. The remaining 35% is unexplained — due to other factors not in the model, measurement error, or inherent randomness. R^2 alone does not confirm the model is appropriate; always check the residuals plot.

Q The p-value shows exactly 0.0000. Is that right?

The toolkit displays p-values to 4 decimal places. If the value rounds to 0.0000, the true p-value is less than 0.00005. This means extremely strong evidence against H0. You can report it as $p < 0.0001$.

Charts and Visualizations

Q How do I save a chart as an image?

Right-click on any chart and select "Save image as" (Chrome/Firefox) or "Copy Image" then paste into a document. Alternatively, use your browser's built-in screenshot tool (Ctrl+Shift+S in Chrome, Cmd+Shift+4 on Mac).

Q The correlation matrix is showing wrong values / looks empty.

The matrix uses the first 6 numeric columns. If your dataset has fewer than 2 numeric columns, the matrix cannot render. Ensure you have loaded a dataset with at least 2 numeric columns.

Q The scatter plot shows no points.

This happens when both selected columns have no complete (non-null) pairs. Check the Explorer tab for missing values in those columns.

Q Can I export charts as PDF or SVG?

Not directly from the toolkit. Chart.js renders to HTML5 canvas elements. For high-resolution exports: right-click and save as PNG, or use browser print (Ctrl+P) to print to PDF, or use a browser extension like "Full Page Screen Capture".

Regression & Advanced

Q Can I run multiple regression (more than one predictor)?

The current version only supports simple linear regression (one predictor). For multiple regression, export your data and use Python (statsmodels, scikit-learn) or R (lm function).

Q The regression line looks wrong — it does not follow the data.

OLS regression minimises the sum of squared vertical residuals. If your data has strong outliers, the line may be pulled toward them. Check the residuals vs fitted plot for extreme residuals. Consider removing outliers using the IQR detection in the Cleaning module and re-fitting.

Q What is the prediction interval vs confidence interval in regression?

The confidence interval for a mean prediction covers where the average Y would be for a given X across many repetitions. The prediction interval (shown in the Predict panel) covers where a single new observation would fall — it is always wider because it includes both estimation uncertainty and individual variability.

Distributions

Q What is the difference between PDF and CDF?

The PDF (probability density function) shows the relative likelihood of each value. For continuous distributions, the PDF value at a point is not itself a probability — it must be integrated over an interval. The CDF at point x gives $P(X \leq x)$, the cumulative probability up to x. The CDF is always between 0 and 1.

Q The Binomial and Poisson distributions show bars instead of a curve. Is that correct?

Yes. Binomial and Poisson are discrete distributions — X can only take integer values (0, 1, 2, ...). Each bar represents the probability mass at that integer. Continuous distributions (Normal, Exponential, Uniform, t) use a smooth line because X can take any value.

Q The KS test says 'Fits Normal' but the Q-Q plot looks curved. Which should I trust?

With small samples ($n < 30$), the KS test has low power and will often fail to reject even when the distribution is non-normal. The Q-Q plot is a more visual and sensitive diagnostic for small samples. If the points curve systematically away from the diagonal, treat the distribution as non-normal regardless of the KS result.

14. Statistical Glossary

Key terms used throughout the toolkit, in alphabetical order.

Alpha (alpha)

The significance level — the probability threshold below which we reject H_0 . Common values: 0.05, 0.01, 0.10.

CDF

Cumulative Distribution Function — $P(X \leq x)$. Always increases from 0 to 1.

Cohen's d

A measure of effect size for the difference between two means, expressed in standard deviation units. $d=0.2$ small, 0.5 medium, 0.8 large.

CV (Coefficient of Variation)

Standard deviation expressed as a percentage of the mean. Useful for comparing variability across variables with different units.

Effect Size

A standardised measure of the practical magnitude of a result, independent of sample size. Examples: Cohen's d , eta-squared.

F-statistic

A ratio of variance estimates. Used in ANOVA and regression. Large F with small p indicates the model explains significant variance.

H_1 (Alternative Hypothesis)

The hypothesis we accept if H_0 is rejected. For a two-tailed test: parameter $\neq$ hypothesised value.

KS Test

Kolmogorov-Smirnov test. Measures the maximum distance between an empirical and a theoretical CDF. Used for goodness-of-fit.

Mean

The arithmetic average. Sum of values divided by n . Sensitive to outliers.

Mode

The most frequently occurring value.

ANOVA

Analysis of Variance — tests whether the means of three or more groups are all equal using the F -statistic.

Central Limit Theorem

States that the distribution of sample means approaches Normal as sample size increases, regardless of the population distribution.

Confidence Interval

A range of plausible values for a population parameter at a given confidence level (e.g., 95% CI for the mean).

Degrees of Freedom (df)

The number of independent pieces of information used to estimate a parameter. For a one-sample t -test, $df = n - 1$.

Empirical CDF

The CDF computed from observed data — the proportion of observations $\leq x$.

H_0 (Null Hypothesis)

The default assumption of no effect or no difference. We test whether there is sufficient evidence to reject it.

IQR (Interquartile Range)

Q_3 minus Q_1 . The range of the middle 50% of the data. Robust to outliers.

Kurtosis

Excess kurtosis measures tail weight relative to a Normal distribution. Positive = heavier tails; negative = lighter tails.

Median

The middle value when data are sorted. The 50th percentile. Robust to outliers.

Normal Distribution

The bell-shaped distribution defined by mean (μ) and standard deviation (σ). Central to many statistical tests.

OLS (Ordinary Least Squares)

Regression method that minimises the sum of squared vertical residuals.

p-value

Probability of observing results at least as extreme as those obtained, assuming H_0 is true. NOT the probability that H_0 is true.

Pearson r

Correlation coefficient measuring linear association between two variables. Range: -1 to +1.

Power

The probability of correctly rejecting H_0 when H_1 is true. Power = 1 - beta. Increases with sample size and effect size.

R-squared (R^2)

Proportion of variance in Y explained by the regression model. Range 0 to 1. Does not indicate whether the model is appropriate.

SEM (Standard Error of Mean)

$SD / \sqrt{n}$. Measures precision of the sample mean as an estimate of the population mean.

Standard Deviation

Average distance of observations from the mean. $\sqrt{\text{variance}}$. Same units as the data.

Variance

Mean squared deviation from the mean. Sum of $(x - \bar{x})^2 / (n-1)$. Units are squared.

Outlier

An observation substantially different from most others. IQR method: outside $Q1 - 1.5 \cdot IQR$ or $Q3 + 1.5 \cdot IQR$.

PDF

Probability Density Function — for continuous distributions, the height of the distribution curve at a value.

Percentile

The value below which a given percentage of observations fall. $Q1 = 25\text{th percentile}$.

Q-Q Plot

Quantile-Quantile plot. Compares sample quantiles to theoretical Normal quantiles. Points on the diagonal = Normal.

Residual

Difference between an observed value and its predicted value: $e = y - \hat{y}$.

Skewness

Asymmetry of the distribution. Positive = right tail; negative = left tail.

t-statistic

A standardised test statistic: $(\text{estimate} - \text{null value}) / \text{standard error}$. Follows a t-distribution under H_0 .

Welch t-test

Two-sample t-test that does not assume equal variances. Uses the Satterthwaite approximation for degrees of freedom. More robust than Student's t-test.

15. Troubleshooting

Issue	Charts do not appear
Cause: Chart.js CDN did not load. Solution: Ensure you have an internet connection on first load, or download Chart.js locally and update the script src in the HTML head to a local path.	
Issue	Fonts look wrong (generic sans-serif)
Cause: Google Fonts CDN did not load. Solution: The toolkit still works — only the font is affected. For offline use, self-host Inter and JetBrains Mono and update the link tag.	
Issue	CSV loads but all columns are categorical
Cause: Your CSV uses a semicolon (;) or pipe () as delimiter instead of comma. Solution: Open in Excel or LibreOffice and re-save as comma-separated CSV.	
Issue	Statistics show NaN
Cause: The selected column has zero non-null numeric values. Solution: Check the Explorer tab — the column may be all-null or classified as categorical due to text values.	
Issue	Regression shows identical results for different X columns
Cause: The X dropdown may not have updated. Solution: Click the X dropdown and re-select to force a refresh.	
Issue	The t-test slider only goes to a small value
Cause: The slider max is set to 1.5x the column maximum on tab load. Solution: If you need to test a value above the data range, switch to the column closest to your target value, or manually edit the HTML if needed.	
Issue	Page freezes after loading a large CSV
Cause: Very large files (>10,000 rows) can overload the browser. Solution: Pre-filter your data in Excel or Python before loading. The toolkit is optimised for datasets up to ~5,000 rows.	

Issue	Goodness-of-fit always says "Fits Normal"
Cause: With small samples ($n < 15$), the KS critical value is large, so the test rarely rejects. Solution: Use the Q-Q plot as the primary diagnostic for small samples.	
Issue	Export downloads an empty file
Cause: The browser blocked the file download. Solution: Check your browser download permissions or try a different browser.	
Issue	Auto-Insights shows no correlation cards
Cause: No pair of numeric columns has $ r > 0.6$. Solution: This is a result, not a bug — your dataset has no strong pairwise linear correlations.	

Data Analysis Toolkit

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Glossary terms

[Open index.html in any modern browser to get started.](#)

No installation. No server. No coding.